


Bihamiltonian FO structures.

based on j. work w. Zheng Hua
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1. FO bracket ass. w. ell. curve C , $\mathcal{V}_{\text{stable}}$

π_C Pois. bracket

on $\mathbb{P}H^0(C, \mathcal{V})^{\vee} = \mathbb{P}^{n-1}$

$\deg \mathcal{V} = n > 0$ v.b.
 $\text{rk} = k$

$e \in \text{Ext}^1(\mathcal{V}, \mathcal{O})$

$$T_e^* \mathbb{P}^{n-1} = \langle e \rangle^{\perp} \subset H^0(C, \mathcal{V})$$

$\downarrow$
 s, t

$$\begin{array}{ccc} & \mathcal{V} & \\ t \uparrow & \nearrow & \nwarrow \tilde{t} \\ 0 & \rightarrow E \rightarrow \mathcal{V} & \\ & \tilde{s} \nearrow & \uparrow s \\ & & 0 \end{array}$$

$$\langle \tilde{t} \circ \tilde{s}, e \rangle =: \langle s, t \rangle$$

$$\pi_C \in H^0(\mathbb{P}^{n-1}, \wedge^2 T)$$

$$[\pi_C, \pi_C] = 0$$

2. $\text{rk} = 1$

Def. π_1 & π_2 on X are compatible

if $\lambda \pi_1 + \mu \pi_2$ is again Poisson

$$\Leftrightarrow [\pi_1, \pi_2] = 0$$

lin. subspaces of Poisson brackets

$$E_1 \subset \mathbb{P}^n \supset E_2 \rightsquigarrow \pi_{E_1}, \pi_{E_2}$$

Thm (Hua-P). S - sm. proj. surface

$(\mathcal{O}_S, \mathcal{V})$ - exc. pair

$$c_1(\mathcal{V}) \cdot (-K_S) > 0$$

$$C \subset S \text{ antican.} \Rightarrow H^0(S, \mathcal{V}) \simeq H^0(C, \mathcal{V}|_C)$$

get $\text{lin. map} \cdot \kappa : H^0(S, -K_S) \rightarrow H^0(\mathbb{P}H^0(\mathcal{V})^*, \wedge^2 T)$

$\text{im } \kappa$ is subspace of comp. Pois. br.

Assume

$$\bullet \quad c_1(\mathcal{V}) \cdot (-K_S) > \text{rk } \mathcal{V}$$

$$\text{then } \ker(\kappa) \subset \text{discr. locus} = \left\{ \begin{array}{l} \text{sing.} \\ \text{antican. div.} \end{array} \right\}$$

Ex. $S = \mathbb{P}^2$ $(\mathcal{O}, \mathcal{V}) \Leftrightarrow \mathcal{V} \in \langle \mathcal{O}(1), \mathcal{O}(2) \rangle$

κ injective by GL_3 -invariance

$$\mathcal{V} = \mathcal{O}(1) \rightsquigarrow \text{all Pois. br. on } \mathbb{P}^2$$

$$\mathcal{V} = \mathcal{O}(2) \rightsquigarrow 10\text{-dim. space of comp. P. b. on } \mathbb{P}^2$$

Ex. $S = \mathbb{P}(\mathcal{O} + \mathcal{O}(n))$ $H^0(S, -K_S)$ is 9 dim.

$$|n| \leq 3 \quad \downarrow_{\mathbb{P}^1} \quad (\mathcal{O}, \mathcal{L}) \quad \text{some line bundle}$$

$\rightsquigarrow$ can get 9 dim. space of comp. P. b. on $\mathbb{P}^{N_V}$
cont. any FO bracket of type

known to

[Odesskii - Wolf]

$(N, 1)$

For which $(n, k), C$ $n > k > 0$

$\exists$ bihamiltonian str. cont. FO-bracket
of type (n, k)

known: • $k=1$

• $n > k, n \equiv \pm 1 \pmod k$
 k odd

Case of $\mathbb{P}^3$ (consider $E \subset \mathbb{P}^3$
deg=4)

$E \subset \mathbb{P}^3 \rightsquigarrow \pi_E$ FO bracket.

Thm (Maurer-P) i) $E_1, E_2 \subset \mathbb{P}^3$

$\{\pi_{E_1}, \pi_{E_2}\} = 0 \Leftrightarrow \exists$ quadric surface Q

2) $(E_i)_{i \in I} \quad E_i \subset \mathbb{P}^3$

$E_1, E_2 \subset Q$

all compatible $\Leftrightarrow$

either $\exists Q$
all $E_i \subset Q$

or $\exists$ 3dim space
of quadric

$E_i = (q_1, q_2) \subset (q_1, q_2)$

$\Leftrightarrow$ already known.

$\Rightarrow$) Note: π_E vanishes on $E \subset \mathbb{P}^3$
 $\mathbb{P}^3 - E \rightarrow \mathbb{P}^1$

key calculation: some bivector on $\mathbb{P}^3$

$$1) p \in E \subset \mathbb{P}^3, \quad [\pi_E, \pi] \Big|_p = 0$$

$$\Rightarrow \pi|_p = v_1 \wedge v_2, \quad v_i \in T_p E$$

$$2) \text{ take } x \in \mathbb{P}^3 - E$$

$$\exists! Q \ni x, Q \supset E$$

$$T_x Q = \text{im}(\pi_E : T_x^* \mathbb{P}^3 \rightarrow T_x \mathbb{P}^3).$$

Proof of Thm. Suppose $[\pi_{E_1}, \pi_{E_2}] = 0$.

$$E_1 \neq E_2$$

$$p \in E_1, \text{ by 1), } T_p E_1 \subset \text{im } \pi_{E_2} \subset T_p \mathbb{P}^3$$

// by 2

$$\Rightarrow E_1 \hookrightarrow \mathbb{P}^3 \xrightarrow{(\pi_{E_2})^*} \mathbb{P}^1$$

has zero deriv.

$$\Rightarrow E_1 \longrightarrow \mathbb{P}^1$$

constant

$$\Rightarrow E_1 \subset Q$$

$\cup$
 E_2

How to calculate $(\pi_E, \pi)|_p$ at $p \in E$?

$\pi_E|_p = 0$, π_E^{lin} can be computed

$$\sum_{\substack{i,j \\ \uparrow \\ \text{lin. functions}}} l_{ij} \alpha_{x_i} \alpha_{x_j}$$

 get Lie alg. str-ure

on $T_p^* \mathbb{P}^3 = m/m^2$ induced by $\{, \}$
 $\{m, m\} \subset m$

have corresponding extension

$$\begin{array}{ccc} 0 \rightarrow \mathcal{V} \xrightarrow{i} \mathcal{V}^{\wedge} \xrightarrow{p} L & & T^* \mathcal{P}^{n-1} \subset H^0(E, L) \\ & & \uparrow \langle e \rangle^{\perp L} \\ \text{End}(\mathcal{V}) / \langle \text{id} \rangle & \xrightarrow{\quad} & H^0(E, L) \\ & \searrow & \uparrow \langle e \rangle^{\perp} \\ A & \xrightarrow{\quad} & p \circ A \circ i \end{array}$$

this induces an isom-ism

$$\text{End}(\mathcal{V}) / \langle \text{id} \rangle \cong \ker(\pi_E : T^* \rightarrow T)$$

comp. w. Lie alg. str-ures

ext-ns corr. to $p \in E$ are of the form

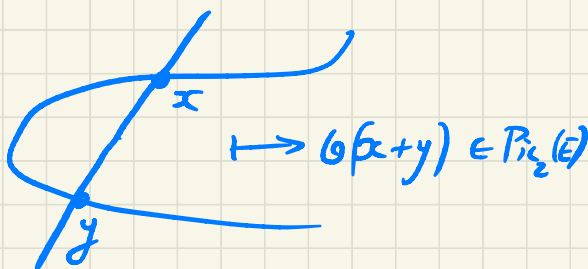
$$0 \rightarrow \mathcal{O}(p) \oplus L(-p) \rightarrow L$$

$$\underbrace{x_1, \dots, x_{n-2}}_{\text{abelian}}, y \quad \{y, x_i\} = x_i$$

$$\Rightarrow \pi_E^{\text{lin}} = \dots$$

Now consider normal ell. curves in $\mathbb{P}^n$, $n \geq 4$

$$\begin{array}{c} E \subset \mathbb{P}^n \\ \uparrow \subset \\ S_L \subset \text{Sec}^2(E) \\ \downarrow \quad \downarrow \\ L \in \text{Pic}_2(E) \end{array}$$



$\leadsto$ family of (S_L)
 $\uparrow$ scrolls $L \in \text{Pic}_2(E)$

Thm. $E_1, E_2 \subset \mathbb{P}^n$, $n \geq 4$

$$[\pi_{E_1}, \pi_{E_2}] = 0 \text{ iff}$$

either $\exists$ scroll S
 $E_1, E_2 \subset S$

or $n = 5$

$$\exists \text{ Veronese}(\mathbb{P}^2) \subset \mathbb{P}^5 \\ \downarrow \\ E_1, E_2$$

+ similar statement
 for (E_i) .

Additional ingredients: for $E \subset \mathbb{P}^n$, $n \geq 4$
 $p \notin E$ $\text{rk } \pi_E|_p = 2 \iff p \in \text{Sec}^2(E) - E$
 $\text{im}(\pi_E^* : T^* \rightarrow T) = T_p S_L$
 $\downarrow$
 $E \subset \text{Ver } \mathbb{P}^2$ $n=5$

